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Inventor(s)	Jia; Zhensheng et al.

Communication systems and methods

Abstract

An injection locked transmitter for an optical communication network includes a master seed laser source input substantially confined to a single longitudinal mode, an input data stream, and a laser injected modulator including at least one slave laser having a resonator frequency that is injection locked to a frequency of the single longitudinal mode of the master seed laser source. The laser injected modulator is configured to receive the master seed laser source input and the input data stream, and output a laser modulated data stream.

Inventors: Jia; Zhensheng (Superior, CO), Campos; Luis Alberto (Superior, CO), Knittle; Curtis Dean (Superior, CO)

Applicant: CABLE TELEVISION LABORATORIES, INC (Louisville, CO)

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Assignee: Cable Television Laboratories, Inc. (Louisville, CO)

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Primary Examiner: Bello; Agustin

Attorney, Agent or Firm: Snider IP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/330,203, filed May 25, 2021, which application is a divisional of U.S. application Ser. No. 16/093,594, filed Oct. 12, 2018. U.S. application Ser. No. 16/093,594 is a National Stage Entry of PCT/US17/23355, filed on Mar. 21, 2017. PCT/US17/23355 is a continuation of U.S. patent application Ser. No. 15/283,632, filed Oct. 3, 2016, now U.S. Pat. No. 9,912,409, issued Mar. 6, 2018. U.S. patent application Ser. No. 15/283,632 claims the benefit of and priority to U.S. Provisional Patent Application Ser. No. 62/321,211, filed Apr. 12, 2016. The respective disclosures of all these previous applications are incorporated herein by reference in their entireties.

BACKGROUND

(1) The field of the disclosure relates generally to fiber communication networks, and more particularly, to optical networks utilizing wavelength division multiplexing.

(2) Telecommunications networks include an access network through which end user subscribers connect to a service provider. Bandwidth requirements for delivering high-speed data and video services through the access network are rapidly increasing to meet growing consumer demands. At present, data delivery over the access network is growing by gigabits (Gb)/second for residential subscribers, and by multi-Gb/s for business subscribers. Present access networks are based on passive optical network (PON) access technologies, which have become the dominant system architecture to meet the growing high capacity demand from end users.

(3) Gigabit PON and Ethernet PON architectures are conventionally known, and presently provide about 2.5 Gb/s data rates for downstream transmission and 1.25 Gb/s for upstream transmission (half of the downstream rate). 10 Gb/s PON (XG-PON or IEEE 10G-EPON) has begun to be implemented for high-bandwidth applications, and a 40 Gb/s PON scheme, which is based on time and wavelength division multiplexing (TWDM and WDM) has recently been standardized. A growing need therefore exists to develop higher/faster data rates per-subscriber to meet future bandwidth demand, and also increase the coverage for services and applications, but while also minimizing the capital and operational expenditures necessary to deliver higher capacity and performance access networks.

(4) One known solution to increase the capacity of a PON is the use of WDM technology to send a dedicated wavelength signal to end users. Current detection scheme WDM technology, however, is limited by its low receiver sensitivity, and also by the few options available to upgrade and scale the technology, particularly with regard to use in conjunction with the lower-quality legacy fiber environment. The legacy fiber environment requires operators to squeeze more capacity out of the existing fiber infrastructure to avoid costs associated with having to retrench new fiber installment. Conventional access networks typically include six fibers per node, servicing as many as 500 end users, such as home subscribers. Conventional nodes cannot be split further and do not typically contain spare (unused) fibers, and thus there is a need to utilize the limited fiber availability in a more efficient and cost-effective manner.

(5) Coherent technology has been proposed as one solution to increase both receiver sensitivity and overall capacity for WDM-PON optical access networks, in both brown and green field deployments. Coherent technology offers superior receiver sensitivity and extended power budget, and high frequency selectivity that provides closely-spaced dense or ultra-dense WDM without the need for narrow band optical filters. Moreover, a multi-dimensional recovered signal experienced by coherent technology provides additional benefits to compensate for linear transmission impairments such as chromatic dispersion (CD) and polarization-mode dispersion (PMD), and to efficiently utilize spectral resources to benefit future network upgrades through the use of multi-

level advanced modulation formats. Long distance transmission using coherent technology, however, requires elaborate post-processing, including signal equalizations and carrier recovery, to adjust for impairments experienced along the transmission pathway, thereby presenting significant challenges by significantly increasing system complexity.

(6) Coherent technology in longhaul optical systems typically requires significant use of high quality discrete photonic and electronic components, such as digital-to-analog converters (DAC), analog-to-digital converters (ADC), and digital signal processing (DSP) circuitry such as an application-specific integrated circuit (ASIC) utilizing CMOS technology, to compensate for noise, frequency drift, and other factors affecting the transmitted channel signals over the long distance optical transmission. Coherent pluggable modules for metro solution have gone through C Form-factor pluggable (CFP) to CFP2 and future CFP4 via multi-source agreement (MSA) standardization to reduce their footprint, to lower costs, and also to lower power dissipation. However, these modules still require significant engineering complexity, expense, size, and power to operate, and therefore have not been efficient or practical to implement in access applications.

BRIEF SUMMARY

(7) In one aspect, an injection locked transmitter for an optical communication network includes a master seed laser source input substantially confined to a single longitudinal mode, an input data stream, and a laser injected modulator including at least one slave laser having a resonator frequency that is injection locked to a frequency of the single longitudinal mode of the master seed laser source. The laser injected modulator is configured to receive the master seed laser source input and the input data stream, and output a laser modulated data stream.

(8) In another aspect, an optical network communication system includes, an input signal source, an optical frequency comb generator configured to receive the input signal source and output a plurality of phase synchronized coherent tone pairs. Each of the plurality of phase synchronized coherent tone pairs includes a first unmodulated signal and a second unmodulated signal. The system further include a first transmitter configured to receive the first unmodulated signal of a selected one of the plurality of phase synchronized coherent tone pairs as a seed source and to output a first modulated data stream, and a first receiver configured to receive the first modulated data stream from the first transmitter and receive the second unmodulated signal of the selected one of the plurality of phase synchronized coherent tone pairs as a local oscillator source.

(9) In yet another aspect, an optical network communication system includes an optical hub including an optical frequency comb generator configured to output at least one phase synchronized coherent tone pair having a first unmodulated signal and a second unmodulated signal, and a downstream transmitter configured to receive the first unmodulated signal as a seed source and to output a downstream modulated data stream. The system further includes a fiber node and an end user including a downstream receiver configured to receive the downstream modulated data stream from the downstream transmitter and receive the second unmodulated signal as a local oscillator source.

(10) In a still further aspect, a method of optical network processing includes steps of generating at least one pair of first and second unmodulated phase synchronized coherent tones, transmitting the first unmodulated phase synchronized coherent tone to a first transmitter as a seed signal, adhering downstream data, in the first transmitter, to the first unmodulated phase synchronized coherent tone to generate a first modulated data stream signal, optically multiplexing the first modulated data stream signal and the second unmodulated phase synchronized coherent tone together within a hub optical multiplexer, and communicating the multiplexed first modulated data stream signal and the second unmodulated phase synchronized coherent tone to a first receiver, by way of fiber optics, for downstream heterodyne detection.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:
- (2) FIG. 1 is a schematic illustration of an exemplary fiber communication system in accordance with an exemplary embodiment of the present disclosure.
- (3) FIG. 2 is a schematic illustration depicting an exemplary transmitter that can be utilized with the fiber communication system depicted in FIG. 1.
- (4) FIG. 3 is a schematic illustration depicting an alternative transmitter that can be utilized with the fiber communication system depicted in FIG. 1.
- (5) FIG. 4 is a schematic illustration depicting an alternative transmitter that can be utilized with the fiber communication system depicted in FIG. 1.
- (6) FIG. 5 is a schematic illustration depicting an alternative transmitter that can be utilized with the fiber communication system depicted in FIG. 1.
- (7) FIG. 6 is a schematic illustration depicting an exemplary upstream connection that can be utilized with the fiber communication system depicted in FIG. 1.
- (8) FIG. 7 is a schematic illustration depicting an exemplary processing architecture implemented with the fiber communication system depicted in FIG. 1.
- (9) FIG. 8 is a flow chart diagram of an exemplary downstream optical network process.
- (10) FIG. 9 is a flow chart diagram of an exemplary upstream optical network process that can be implemented with the downstream process depicted in FIG. 8.
- (11) Unless otherwise indicated, the drawings provided herein are meant to illustrate features of embodiments of this disclosure. These features are believed to be applicable in a wide variety of systems including one or more embodiments of this disclosure. As such, the drawings are not meant to include all conventional features known by those of ordinary skill in the art to be required for the practice of the embodiments disclosed herein.

DETAILED DESCRIPTION

- (12) In the following specification and the claims, reference will be made to a number of terms, which shall be defined to have the following meanings.
- (13) The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.
- (14) “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where the event occurs and instances where it does not.
- (15) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Here and throughout the specification and claims, range limitations may be combined and/or interchanged; such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise.
- (16) FIG. 1 is a schematic illustration of an exemplary fiber communication system **100** in accordance with an exemplary embodiment of the present disclosure. System **100** includes an optical hub **102**, a fiber node **104**, and an end user **106**. Optical hub **102** is, for example, a central office, a communications hub, or an optical line terminal (OLT). In the embodiment shown, fiber node **104** is illustrated for use with a passive optical network (PON). End user **106** is a downstream termination unit, which can represent, for example, a customer device, customer premises (e.g., an apartment building), a business user, or an optical network unit (ONU). In an exemplary

embodiment, system **100** utilizes a coherent Dense Wavelength Division Multiplexing (DWDM) PON architecture.

(17) Optical hub **102** communicates with fiber node **104** by way of downstream fiber **108**.

Optionally, where upstream communication is desired along system **100**, optical hub **102** further connects with fiber node **104** by way of upstream fiber **110**. In operation, downstream fiber **108** and upstream fiber **110** are typically 30 km or shorter. However, according to the embodiments presented herein, greater lengths are contemplated, such as between 100 km and 1000 km. In an exemplary embodiment, fiber node **104** connects with end user **106** by way of fiber optics **112**. Alternatively, fiber node **104** and end user **106** may be integrated as a single device, such as a virtualized cable modem termination system (vCMTS), which may be located at a customer premises. Where fiber node **104** and end user **106** are separate devices, fiber optics **112** typically spans a distance of approximately 5000 feet or less.

(18) Optical hub **102** includes an optical frequency comb generator **114**, which is configured to receive a high quality source signal **116** from an external laser **118** and thereby generate multiple coherent tones **120(1)**, **120(1')**, . . . **120(N)**, **120(N')**. Optical frequency comb generator **114** utilizes, for example, a mode-locked laser, a gain-switched laser, or electro-optic modulation, and is constructed such that multiple coherent tones **120** are generated as simultaneous low-linewidth wavelength channels of known and controllable spacing. This advantageous aspect of the upstream input signal into system **100** allows a simplified architecture throughout the entire downstream portion of system **100**, as described further below.

(19) Generated coherent tones **120** are fed into an amplifier **122**, and the amplified signal therefrom is input into a first hub optical demultiplexer **124**. In an exemplary embodiment, amplifier **122** is an erbium-doped fiber amplifier (EDFA). Optical hub **102** further includes a downstream transmitter **126** and a hub optical multiplexer **128**. In an embodiment, optical hub **102** optionally includes a hub optical splitter **130**, an upstream receiver **132**, and a second hub optical demultiplexer **134**.

(20) Downstream transmitter **126** includes a downstream optical circulator **136** and a downstream modulator **138**. In an exemplary embodiment, downstream modulator **138** is an injection locked laser modulator. Upstream receiver **132** includes an upstream integrated coherent receiver (ICR) **140**, an upstream analog to digital converter (ADC) **142**, and an upstream digital signal processor (DSP) **144**. In the exemplary embodiment, fiber node **104** includes a node optical demultiplexer **146**. In an alternative embodiment, where upstream transmission is desired, fiber node **104** further includes a node optical multiplexer **148**. In the exemplary embodiment, node optical demultiplexer **146** and node optical multiplexer **148** are passive devices.

(21) End user **106** further includes a downstream receiver **150**. In an exemplary embodiment, downstream receiver **150** has a similar architecture to upstream receiver **132**, and includes a downstream ICR **152**, a downstream ADC **154**, and a downstream DSP **156**. For upstream transmission, end user **106** optionally includes end user optical splitter **158**, which may be located within downstream receiver **150** or separately, and an upstream transmitter **160**. In an exemplary embodiment, upstream transmitter **160** has a similar architecture to downstream transmitter **126**, and includes an upstream optical circulator **162**, and an upstream modulator **164**.

(22) In operation, system **100** utilizes optical frequency comb generator **114** and amplifier **122** convert the input high quality source signal **116** into multiple coherent tones **120** (e.g., 32 tones, 64 tones, etc.), which are then input to first hub optical demultiplexer **124**. In an exemplary embodiment, high quality source signal **116** is of sufficient amplitude and a narrow bandwidth such that a selected longitudinal mode of signal **116** is transmitted into optical frequency comb generator **114** without adjacent longitudinal modes, which are suppressed prior to processing by comb generator **114**. First hub optical demultiplexer **124** then outputs a plurality of phase synchronized coherent tone pairs **166(1)**, **166(2)**, . . . **166(N)**. That is, the generated coherent frequency tones **120** are amplified by amplifier **122** to enhance optical power, and then demultiplexed into multiple separate individual phased synchronized coherent tone source pairs **166**. For simplicity of

discussion, the following description pertains only to coherent tone pair **166(1)** corresponding to the synchronized pair signal for the first channel output, which includes a first unmodulated signal **168** for Ch1 and a second unmodulated signal **170** for Ch1', and their routing through system **100**.

(23) With source signal **116** of a high quality, narrow band, and substantially within a single longitudinal mode, coherent tone pair **166(1)**, including first unmodulated signal **168** (Ch1) and second unmodulated signal **170** (Ch1'), is output as a high quality, narrowband signal, which then serves as both a source of seed and local oscillator (LO) signals for both downstream and upstream transmission and reception directions of system **100**. That is, by an exemplary configuration, the architecture of optical frequency comb generator **114** advantageously produces high quality continuous wave (CW) signals. Specifically, first unmodulated signal **168** (Ch1) may function as a downstream seed and upstream LO throughout system **100**, while second unmodulated signal **170** (Ch1') concurrently may function as an upstream seed and downstream LO for system **100**.

(24) According to the exemplary embodiment, within optical hub **102**, first unmodulated signal **168** (Ch1) is divided by hub optical splitter **130** and is separately input to both downstream transmitter **126** and upstream receiver **132** as a "pure" signal, and i.e., substantially low amplitude, narrow bandwidth continuous wave does not include adhered data. First unmodulated signal **168** (Ch1) thus becomes a seed signal for downstream transmitter **126** and an LO signal for upstream receiver **132**. In an exemplary embodiment, within downstream transmitter **126**, first unmodulated signal **168** (Ch1) passes through downstream optical circulator **136** into downstream modulator **138**, in which one or more laser diodes (not shown in FIG. 1, described below with respect to FIGS. 2-5) are excited, and adhere data (also not shown in FIG. 1, described below with respect to FIGS. 2-5) to the signal that then exits downstream optical circulator **136** as downstream modulated data stream **172** (Ch1).

(25) In an exemplary embodiment, downstream optical circulator **136** is within downstream transmitter **126**. Alternatively, downstream optical circulator **136** may be physically located separately from downstream transmitter **126**, or else within the confines of downstream modulator **138**. Downstream modulated data stream **172** (Ch1) is then combined in hub optical multiplexer **128** with the plurality of modulated/unmodulated data stream pairs from other channels (not shown) and transmitted over downstream fiber **108**, to a node optical demultiplexer **174** in fiber node **104**, which then separates the different channel stream pairs for transmission to different respective end users **106**. At end user **106**, because the data stream pair **170**, **172** entering downstream receiver **150** is a phase synchronized, digital signal processing at downstream DSP **156** is greatly simplified, as described below with respect to FIG. 7.

(26) Where upstream reception is optionally sought at optical hub **102**, second unmodulated signal **170** (Ch1') is divided, within end user **106**, by end user optical splitter **158** and is separately input to both downstream receiver **150** and upstream transmitter **160** as a "pure" unmodulated signal for Ch1'. In this alternative embodiment, second unmodulated signal **170** (Ch1') thus functions a seed signal for upstream transmitter **160** and a "pseudo LO signal" for downstream receiver **150** for the coherent detection of Ch1. For purposes of this discussion, second unmodulated signal **170** (Ch1') is referred to as a "pseudo LO signal" because it uses an LO signal from a remote source (output from first hub optical demultiplexer **124**), and is not required to produce an LO signal locally at end user **106**. This particular configuration further significantly reduces cost and complexity of the architecture of the system **100** by the reduction of necessary electronic components.

(27) For upstream transmission, in an exemplary embodiment, a similar coherent detection scheme is implemented for upstream transmitter **160** as is utilized for downstream transmitter **126**. That is, second unmodulated signal **170** (Ch1') is input to upstream optical circulator **162** and modulated by upstream modulator **164** to adhere symmetric or asymmetric data (not shown, described below with respect to FIG. 6) utilizing one or more slave lasers (also not shown, described below with respect to FIG. 6), and then output as an upstream modulated data stream **176** (Ch1'), which is then combined with similar modulated data streams from other channels (not shown) by a node

multiplexer **178** in fiber node **104**. Second unmodulated signal **170** (Ch1') is then transmitted upstream over upstream fiber **110**, separated from other channel signals by second hub optical demultiplexer **134**, an input to upstream receiver **132**, for simplified digital signal processing similar to the process described above with respect to downstream receiver **150**.

(28) By this exemplary configuration, multiple upstream channels from different end users **106** can be multiplexed at fiber node **104** (or a remote node) and sent back to optical hub **102**. Thus, within optical hub **102**, the same coherent detection scheme may be used at upstream receiver **132** as is used with downstream receiver **150**, except that upstream receiver **132** utilizes first unmodulated signal **168** (Ch1) as the LO and upstream modulated data stream **176** (Ch1') to carry data, whereas downstream receiver **150** utilizes the data stream pair (Ch1, Ch1') in reverse. That is, downstream receiver **150** utilizes second unmodulated signal **170** (Ch1') as the LO and downstream modulated data stream **172** (Ch1) to carry data.

(29) Implementation of the embodiments described herein are useful for migrating hybrid fiber-coaxial (HFC) architectures towards other types of fiber architectures, as well as deeper fiber architectures. Typical HFC architectures tend to have very few fiber strands available from fiber node to hub (e.g. fibers **108**, **110**), but many fiber strands could be deployed to cover the shorter distances that are typical from legacy HFC nodes to end users (e.g., fiber optics **112**). In the exemplary embodiments described herein, two fibers (i.e., fibers **108**, **110**) are illustrated between optical hub **102** and fiber node **104**, which can be a legacy HFC fiber node. That is, one fiber (i.e., downstream fiber **108**) is utilized for downstream signal and upstream seed/downstream LO, and another fiber (i.e., upstream fiber **110**) is utilized for upstream signal. Additionally, three fibers (i.e., fiber optics **112A-C**) are illustrated for each end user from fiber node **104** (e.g., legacy HFC fiber node) to end user **106**. By utilization of the advantageous configurations herein, fiber deeper or all-fiber migration schemes can utilize an HFC fiber node as an optical fiber distribution node, thereby greatly minimizing the need for fiber retrenching from an HFC node to an optical hub.

(30) The architecture described herein, by avoiding the need for conventional compensation hardware, can therefore be structured as a significantly less expensive and more compact physical device than conventional devices. This novel and advantageous system and subsystem arrangement allows for multi-wavelength emission with simplicity, reliability, and low cost. Implementation of optical frequency comb generator **114**, with high quality input source signal **116**, further allows simultaneous control of multiple sources that are not realized by conventional discrete lasers. According to the embodiments herein, channel spacing, for example, may be 25 GHz, 12.5 GHz, or 6.25 GHz, based on available signal bandwidth occupancy.

(31) The embodiments described herein realize still further advantages by utilizing a comb generator (i.e., optical frequency comb generator **114**) that maintains a constant wavelength spacing, thereby avoiding optical beat interference (OBI) that may be prevalent in cases with simultaneous transmissions over a single fiber. In the exemplary embodiment illustrated in FIG. **1**, fiber node **104** is shown as a passive system, and is thus expected to maintain a higher reliability than other migration approaches. Nevertheless, one of ordinary skill in the art, after reading and comprehending present application, will understand how the embodiments disclosed herein may also be adapted to a remote PHY solution, or to a remote cable modem termination system (CMTS) that is included in the fiber node.

(32) As illustrated and described herein, system **100** may utilize an architecture of coherent DWDM-PON incorporate novel solutions to meet the unique requirements of access environment, but with cost-efficient structures not seen in conventional hardware systems. Optical frequency comb generator **114** produces a plurality of simultaneous narrow width wavelength channels with controlled spacing, thereby allowing simplified tuning of the entire wavelength comb. This centralized comb light source in optical hub **102** therefore provides master seeding sources and LO signals for both downstream and upstream directions in heterodyne detection configurations in order to reuse the optical sources throughout the entirety of system **100**. This advantageous

configuration realizes significant cost savings and reduction in hardware complexity over intradyne detection schemes in long-haul systems, for example.

(33) FIG. 2 is a schematic illustration depicting an exemplary downstream transmitter **200** that can be utilized with fiber communication system **100**, depicted in FIG. 1. Downstream transmitter **200** includes downstream optical circulator **136** (see FIG. 1, above) in two-way communication with a laser injected modulator **202**, which includes a laser diode **204**, which receives data **206** from an external data source **208**. In an alternative embodiment, downstream transmitter **200** may include two separate fiber receivers (not shown), which would substitute, and eliminate the need, for downstream optical circulator **136** in the structural configuration shown.

(34) In operation, downstream transmitter **200** performs the same general functions as downstream transmitter **126** (FIG. 1, described above). Laser injected modulator **202** utilizes laser diode **204** as a “slave laser.” That is, laser diode **204** is injection locked by external laser **118**, which functions as a single frequency or longitudinal mode master, or seed, laser to keep the frequency of a resonator mode of laser diode **204** close enough to the frequency of the master laser (i.e., laser **118**) to allow for frequency locking. The principle of downstream transmitter **200** is also referred to as “laser cloning,” where a single high quality master laser (i.e., laser **118**) transmits a narrow bandwidth, low noise signal (i.e., source signal **116**), and a relatively inexpensive slave laser (e.g., laser diode **204**) can be used throughout system **100** to transmit data modulated signals, such as downstream modulated data stream **172** (Ch1). In an exemplary embodiment, laser diode **204** is a Fabry Perot laser diode (FP LD), or a vertical-cavity surface-emitting laser (VCSEL), in comparison with the considerably more expensive distributed feedback laser diodes (DFB LD) that are conventionally used. In an alternative embodiment, laser diode **204** is an LED, which can perform as a sufficient slave laser source according to the embodiments herein due to the utilization of the high quality source signal **116** that is consistently utilized throughout system **100**.

(35) More specifically, first unmodulated signal **168** (Ch1) exiting hub optical splitter **130** is input to downstream optical circulator **136**, which then excites laser diode **204**, that is, laser diode **204** emits light at a specified modulation rate. Laser injected modulator **202** adheres data **206** to the excited Ch1 signal, and the resultant modulated Ch1 signal with adhered data is output from downstream optical circulator **136** as downstream modulated data stream **172** (Ch1). According to this exemplary embodiment, first unmodulated signal **168** (Ch1) is input to downstream transmitter **126** as an unmodulated, low amplitude, narrow bandwidth, low noise “pure” source, and is modulated by laser diode **204**, which is a high amplitude, wide bandwidth device, and resultant downstream modulated data stream **172** (Ch1) is a high amplitude, narrow bandwidth, low noise “pure” signal that can be transmitted throughout system **100** without the need for further conventional compensation means (hardware and programming) Suppression of adjacent longitudinal modes from laser diode **204**, for example, is not necessary because of the exciting source signal (i.e., signal **168**) is of such high quality and narrow bandwidth that output downstream modulated data stream **172** (Ch1) is substantially amplified only within the narrow bandwidth of external laser **118**. In the exemplary embodiment illustrated in FIG. 2, laser injected modulator **202** implements direct modulation.

(36) Optical injection locking as described herein thus improves upon the performance of the relatively less expensive, multi-longitudinal slave laser source (i.e., laser diode **204**) in terms of spectral bandwidth and noise properties. With respect to heterodyne coherent detection, incoming signals (upstream or downstream) can be combined with the LO or pseudo-LO and brought to an intermediate frequency (IF) for electronic processing. According to this exemplary configuration, part of the LO/pseudo-LO optical power can also be employed as the master/seed laser for the reverse transmission direction, at both optical hub **102**, and at end user **106** (described below with respect to FIG. 6), and thus a fully coherent system having a master seed and LO delivery from an optical hub can be achieved in a relatively cost-effective manner comparison with conventional systems.

(37) FIG. 3 is a schematic illustration depicting an alternative downstream transmitter **300** that can be utilized with fiber communication system **100**, depicted in FIG. 1. Downstream transmitter **300** is similar to downstream transmitter **200** (FIG. 2), including the implementation of direct modulation, except that downstream transmitter **300** alternatively utilizes polarization division multiplexing to modulate the Ch1 signal into downstream modulated data stream **172** (Ch1).

(38) Downstream transmitter **300** includes downstream optical circulator **136** (see FIG. 1, above) in two-way communication with a laser injected modulator **302**, which includes a polarization beam splitter (PBS)/polarization beam combiner (PBC) **304**, which can be a single device. Laser injected modulator **302** further includes a first laser diode **306** configured to receive first data **308** from an external data source (not shown in FIG. 3), and a second laser diode **310** configured to receive second data **312** from the same, or different, external data source.

(39) In operation, downstream transmitter **300** is similar to downstream transmitter **200** with respect to the implementation of direct modulation, and master/slave laser injection locking. Downstream transmitter **300** though, alternatively implements dual-polarization from the splitter portion of PBS/PBC **304**, which splits first unmodulated signal **168** (Ch1) into its x-polarization component P1 and y-polarization component P2, which separately excite first laser diode **306** and second laser diode **310**, respectively. Similar to downstream transmitter **200** (FIG. 2), in downstream transmitter **300**, first unmodulated signal **168** (Ch1) exiting hub optical splitter **130** is input to downstream optical circulator **136**, the separate polarization components of which then excite laser diodes **306**, **310**, respectively, at the specified modulation rate. Laser injected modulator **302** adheres data first and second data **308**, **312** to the respective excited polarization components of the Ch1 signal, which are combined by the combiner portion of PBS/PBC **304**. The resultant modulated Ch1 signal with adhered data is output from downstream optical circulator **136** as downstream modulated data stream **172** (Ch1).

(40) In an exemplary embodiment, the polarized light components received by first and second laser diodes **306**, **310** are orthogonal (90 degrees and/or noninteractive). That is, first laser diode **306** and second laser diode **310** are optimized as slave lasers to lock onto the same wavelength as external laser **118** (master), but with perpendicular polarization directions. By this configuration, large data packets (e.g., first data **308** and second data **312**) can be split and simultaneously sent along separate pathways before recombination as downstream modulated data stream **172** (Ch1). Alternatively, first data **308** and second data **312** may come from two (or more) separate unrelated sources. The orthogonal split prevents data interference between the polarized signal components. However, one of ordinary skill in the art will appreciate that, according to the embodiment of FIG. 3, first unmodulated signal **168** (Ch1) can also be polarized at 60 degrees, utilizing similar principles of amplitude and phase, as well as wavelength division. First unmodulated signal **168** (Ch1) can alternatively be multiplexed according to a spiral or vortex polarization, or orbital angular momentum. Additionally, whereas the illustrated embodiment features polarization multiplexing, space division multiplexing and mode division multiplexing may be also alternatively implemented.

(41) According to this exemplary embodiment, master continuous wave signal for Ch1, namely, first unmodulated signal **168**, is received from optical frequency comb generator **114** and is split to be used, in the first part, as the LO for upstream receiver **132**, and in the second part, to synchronize two slave lasers (i.e., first laser diode **306** and second laser diode **310**) by the respective x-polarization and y-polarization light portions such that both slave lasers oscillate according to the wavelength of the master laser (i.e., external laser **118**). Data (i.e., first data **308** and second data **312**) is directly modulated onto the two slave lasers, respectively. This injection locking technique thus further allows for frequency modulation (FM) noise spectrum control from the master laser to the slave laser, and is further able to realize significant improvements in FM noise/phase jitter suppression and emission linewidth reduction.

(42) As described herein, utilization of optical injection with a dual-polarization optical transmitter

(i.e., downstream transmitter **300**) by direct modulation may advantageously implement relatively lower-cost lasers to perform the functions of conventional lasers that are considerably more costly. According to this configuration of a dual-polarization optical transmitter by direct modulation of semiconductor laser together with coherent detection, the present embodiments are particularly useful for short-reach applications in terms of its lower cost and architectural compactness. Similar advantages may be realized for long reach applications.

(43) FIG. 4 is a schematic illustration depicting an alternative downstream transmitter **400** that can be utilized with fiber communication system **100**, depicted in FIG. 1. Downstream transmitter **400** is similar to downstream transmitter **200** (FIG. 2), except that downstream transmitter **400** alternatively implements external modulation, as opposed to direct modulation, to modulate the Ch1 signal into downstream modulated data stream **172** (Ch1). Downstream transmitter **400** includes downstream optical circulator **136** (see FIG. 1, above) and a laser injected modulator **402**. Downstream optical circulator **136** is in one-way direct communication with a separate external optical circulator **404** that may be contained within laser injected modulator **402** or separate. Laser injected modulator **402** further includes a laser diode **406**, which receives the low amplitude, narrow bandwidth, first unmodulated signal **168** (Ch1) and emits an excited, high amplitude, narrow bandwidth, optical signal **408** back to external optical circulator **404**. Laser injected modulator **402** still further includes an external modulating element **410**, which receives data **412** from an external data source **414**, and adheres data **412** with optical signal **408** to be unidirectionally received back by downstream optical circulator **136** and output as downstream modulated data stream **172** (Ch1).

(44) In this exemplary embodiment, downstream transmitter **400** performs the same general functions as downstream transmitter **126** (FIG. 1, described above), but uses external modulation as the injection locking mechanism to lock laser diode **406** to the wavelength of the master laser source (e.g., external laser **118**). To implement external modulation, this embodiment regulates optical signal flow through mostly unidirectional optical circulators (i.e., downstream optical circulator **136**, external optical circulator **404**). External modulating element **410** may optionally include a demultiplexing filter (not shown) as an integral component, or separately along the signal path of downstream modulated data stream **172** (Ch1) prior to input by downstream receiver **150**. In an exemplary embodiment, external modulating element **410** is a monitor photodiode, and injection locking is performed through a rear laser facet.

(45) FIG. 5 is a schematic illustration depicting an alternative downstream **500** transmitter that can be utilized with fiber communication system **100**, depicted in FIG. 1. Downstream transmitter **500** is similar to downstream transmitter **300** (FIG. 3), including the implementation of direct modulation and polarization division multiplexing, except that downstream transmitter **500** further implements quadrature amplitude modulation (QAM) to modulate the Ch1 signal into downstream modulated data stream **172** (Ch1). That is, further external modulating elements may be utilized per polarization branch (FIG. 2, above) to generate QAM signals.

(46) Downstream transmitter **500** includes downstream optical circulator **136** (see FIG. 1, above) in two-way communication with a laser injected modulator **502**, which includes a PBS/PBC **504**, which can be a single device or two separate devices. Additionally, all of the components of laser injected modulator **502** may themselves be separate devices, or alternatively all contained within a single photonic chip. Laser injected modulator **502** further includes a first laser diode **506** configured to receive first data **508** from an external data source (not shown in FIG. 5), a second laser diode **510** configured to receive second data **512** from the same, or different, external data source, a third laser diode **514** configured to receive third data **516** from the same/different, external data source, and a fourth laser diode **518** configured to receive fourth data **520** from the same/different external data source.

(47) In operation, downstream transmitter **500** implements dual-polarization from the splitter portion of PBS/PBC **504**, which splits first unmodulated signal **168** (Ch1) into its x-polarization

component (P1) and y-polarization component (P2). Each polarization component P1, P2 is then input to first non-polarized optical splitter/combiner **522** and second non-polarized optical splitter/combiner **524**, respectively. First and second optical splitters/combiners **522**, **524** each then further split their respective polarization components P1, P2 into their I-signals **526**, **528**, respectively, and also into their Q-signals **530**, **532**, respectively. Generated I-signals **526**, **528** then directly excite laser diodes **506**, **514**, respectively. Before directly communicating with laser diodes **510**, **518**, respectively, generated Q-signals **530**, **532** first pass through first and second quadrature phase shift elements **534**, **536**, respectively, each of which shifts the Q-signal by 45 degrees in each direction, such that the respective Q-signal is offset by 90 degrees from its respective I-signal when recombined at splitters/combiners **522**, **524**.

(48) The resultant modulated Ch1 signal, with adhered data, is output from downstream optical circulator **136** of downstream transmitter **500** as downstream modulated data stream **172** (Ch1), and as a polarized, multiplexed QAM signal. According to this exemplary embodiment, utilization of a photonic integrated circuit allows for directly modulated polarization of a multiplexed coherent system, but utilizing significantly lower cost hardware configurations than are realized by conventional architectures. In an exemplary embodiment, laser diodes **506**, **510**, **514**, **516** are PAM-4 modulated laser diodes capable of generating 16-QAM polarization multiplexed signals.

(49) FIG. **6** is a schematic illustration depicting an exemplary upstream transmitter **600** that can be utilized with the fiber communication system **100**, depicted in FIG. **1**. In the embodiment illustrated in FIG. **6**, upstream transmitter **600** is similar to downstream transmitter **300** (FIG. **3**) in structure and function. Specifically, upstream transmitter **600** includes upstream optical circulator **162** (see FIG. **1**, above) in two-way communication with a laser injected modulator **602** (not separately illustrated in FIG. **6**), which includes a PBS/PBC **604**, which can be a single device or separate devices. Laser injected modulator **602** further includes a first laser diode **606** configured to receive first data **608** from an external data source (not shown in FIG. **6**), and a second laser diode **610** configured to receive second data **612** from the same, or different, external data source. Similar to the embodiments of FIGS. **2-5**, above, downstream transmitter **600** may also eliminate for upstream optical circulator **162** by the utilization of at least two separate fiber receivers (not shown).

(50) Upstream transmitter **600** is thus nearly identical to downstream transmitter **300** (FIG. **3**), except that upstream transmitter **600** utilizes second unmodulated signal **170** (Ch1') as the end user seed source, in laser injected modulator **602**, to combine or adhere with data (e.g., first data **608**, second data **612**) to generate upstream modulated data stream **176** (Ch1') to carry upstream data signals to an upstream receiver (e.g., upstream receiver **132**). In operation, first laser diode **606** and second laser diode **610** also function as slave lasers by injection locking to the master signal from external laser **118**. That is, symmetric or asymmetric data for Ch1' (e.g., first data **608**, second data **612**) is modulated onto the two slave lasers (i.e., first laser diode **606** and second laser diode **610**) with polarization multiplexing, much the same as the process implemented with respect to downstream transmitter **300** (FIG. **3**) in optical hub **102**.

(51) In this example, upstream transmitter **600** is illustrated to substantially mimic the architecture of downstream transmitter **300** (FIG. **3**). Alternatively, upstream transmitter **600** could equivalently mimic the architecture of one or more of downstream transmitters **200** (FIG. **2**), **400** (FIG. **4**), or **500** (FIG. **5**) without departing from the scope of the present disclosure. Furthermore, upstream transmitter **600** can conform to any of the embodiments disclosed by FIGS. **2-5**, irrespective of the specific architecture of the particular downstream transmitter utilized within optical hub **102**. By utilization of high-quality, narrow bandwidth, low noise external laser source **118**, the master/slave laser relationship carries through the entirety of system **100**, and the plurality of end users **106** that receive modulated/unmodulated signal pairs (which may be 32, 64, 128, or as many as 256 from a single fiber line pair, e.g., downstream fiber **108** and upstream fiber **110**).

(52) The significant cost savings according to the present embodiments are thus best realized when

considering that as many as 512 downstream transmitters (e.g., downstream transmitter **126**, FIG. **1**) and upstream transmitters (e.g., upstream transmitter **160**, FIG. **1**) may be necessary to fully implement all available chattel pairs from a single optical hub **102**. The present embodiments implement a significantly lower cost and less complex hardware architecture to utilize the benefits accruing from implementation of high-quality external laser **118**, without having to add expensive single longitudinal mode laser diodes, or other compensation hardware necessary to suppress adjacent longitudinal modes from inexpensive lasers or the noise components produced thereby.

(53) FIG. **7** is a schematic illustration depicting an exemplary processing architecture which can be implemented for upstream receiver **132**, downstream receiver **150**, and fiber communication system **100**, depicted in FIG. **1**. The respective architectures of upstream receiver **132** and downstream receiver **150** are similar with respect to form and function (described above with respect to FIG. **1**), except that upstream receiver **132** receives a first data stream pair **700** for Ch1, Ch1', in reverse of a second data stream pair **702**, which is received by downstream receiver **150**. In other words, as described above, first data stream pair **700** includes first unmodulated signal **168** (Ch1) as the LO and upstream modulated data stream **176** (Ch1') to carry data, whereas second data stream pair **702** includes unmodulated signal **170** (Ch1') as the LO and downstream modulated data stream **172** (Ch1) to carry data.

(54) First and second data stream pairs **700**, **702** the multiplexed phase synchronized pairs modulated/unmodulated of optical signals that are converted into analog electrical signals by ICR **140** and ICR **152**, respectively. The respective analog signals are then converted into digital domain by ADC **142** and ADC **154**, for digital signal processing by DSP **144** and DSP **156**. In an exemplary embodiment, digital signal processing may be performed by a CMOS ASIC employing very large quantities of gate arrays. A conventional CMOS ASIC, for example, can utilize as many as 70 million gates to process incoming digitized data streams. In the conventional systems, modulated data streams for Ch1 and Ch1' are processed independently, which requires significant resources to estimate frequency offset, drift, and digital down conversion compensation factors (e.g., $e^{j\omega t}$, where ω represents the frequency difference between first unmodulated signal **168** and upstream modulated data stream **176**, and to is held constant for coherent tone pair **166**, as extended throughout system **100**).

(55) According to the exemplary embodiments disclosed herein, on the other hand, the modulated and unmodulated signals from Ch1 and Ch1' are phase synchronized together such that the difference between ω of the signal pair is always known, and phase synchronized to maintain a constant relationship. In contrast, conventional systems are required to constantly estimate the carrier phase to compensate for factors such as drift, which requires considerable processing resources, as discussed above. According to the present embodiments though, since Ch1 and Ch1' are synchronized together as first and second data stream pairs **700**, **702**, the offset w between the pairs **700**, **702** need not be estimated, since it may be instead easily derived by a simplified subtraction process in DSP **144** and DSP **156** because the signal pairs will drift together by the same amount in a constant relationship. By this advantageous configuration and process, digital signal processing by a CMOS ASIC can be performed utilizing as few as one million gates, thereby greatly improving the processing speed of the respective DSP, and/or reducing the number of physical chips required to perform the processing (or similarly increasing the amount of separate processing that may be performed by the same chip). At present, implementation of the embodiments described herein may improve downstream and upstream data transmission speeds by as much as 5000 times faster than conventional systems.

(56) FIG. **8** is a flow chart diagram of an exemplary downstream optical network process **800** that can be implemented with fiber communication system **100**, depicted in FIG. **1**. Process **800** begins at step **802**. In step **802**, coherent tone pairs **166** are generated and output by optical frequency comb generator **114**, amplifier **122**, and first hub optical demultiplexer **124**. Similar to the discussion above, for simplification purposes, the following discussion addresses specific coherent

tone pair **166(1)** for Ch1, Ch1'. Coherent tone pair **166** includes first unmodulated signal **168** (Ch1) and second unmodulated signal **170** (Ch1'). Once coherent tone pair **166** is generated, process **800** proceeds from step **802** to steps **804** and **806**, which may be performed together or simultaneously. (57) In step **804**, first unmodulated signal **168** (Ch1) is input to an optical splitter, e.g., optical splitter **130**, FIG. **1**. In step **806**, second unmodulated signal **170** (Ch1') is transmitted to a multiplexer, e.g., hub optical multiplexer **128**, FIG. **1**. Referring back to step **804**, first unmodulated signal **168** (Ch1) is split to function both as an LO for upstream detection, and as a seed for downstream data transmission. For upstream detection, step **804** proceeds to step **808**, where first unmodulated signal **168** (Ch1) is received by an upstream receiver, i.e., upstream receiver **132**, FIG. **1**. For downstream data transmission, step **804** separately and simultaneously proceeds to step **810**.

(58) Step **810** is an optional step, where polarization division multiplexing is desired. In step **810**, first unmodulated signal **168** (Ch1) is split into its x-component and y-component parts P1, P2, respectively (e.g., by PBS/PBC **304**, FIG. **3** or PBS/PBC **504**, FIG. **5**) for separate direct or external modulation. Where polarization division multiplexing is not utilized, process **800** skips step **810**, and instead proceeds directly from step **804** to step **812**. In step **812**, first unmodulated signal **168** (Ch1), or its polarized components if optional step **810** is implemented, is modulated by direct (e.g., FIGS. **2**, **3**, **5**) or external (e.g., FIG. **4**) modulation. Process **800** then proceeds from step **812** to step **814**. Step **814** is an optional step, which is implemented if optional step **810** is also implemented for polarization division multiplexing. In step **814**, the x-component and y-component parts P1, P2 are recombined (e.g., by PBS/PBC **304**, FIG. **3** or PBS/PBC **504**, FIG. **5**) for output as downstream modulated data stream **172** (Ch1). Where polarization division multiplexing was not utilized, process **800** skips step **814**, and instead proceeds directly from step **812** to step **816**.

(59) In step **816**, second unmodulated signal **170** (Ch1') and downstream modulated data stream **172** (Ch1) are optically multiplexed, i.e., by hub optical multiplexer **128**, FIG. **1**, as a phase synchronized data stream pair (e.g., second data stream pair **702**, FIG. **7**). Process **800** then proceeds from step **816** to step **818**, where the phase synchronized data stream pair is transmitted over an optical fiber, i.e., downstream fiber **108**, FIG. **1**. Process **800** then proceeds from step **818** to step **820**, where the synchronized data stream pair is optically demultiplexed, e.g., by node optical demultiplexer **174** in fiber node **104**. Process **800** then proceeds from step **820** to step **822**, where both components of the demultiplexed data stream pair (e.g., second unmodulated signal **170** (Ch1') and downstream modulated data stream **172** (Ch1)) are received by a downstream receiver (e.g., downstream receiver **150**, FIG. **1**) for heterodyne coherent detection.

(60) Where an end user (e.g., end user **106**) further includes upstream transmission capability, process **800** further includes optional steps **824** and **826**. In step **824**, and prior to downstream reception in step **822**, second unmodulated signal **170** (Ch1') is optically split (e.g., by end user optical splitter **158**, FIG. **1**), and additionally transmitted, in step **826**, to an upstream transmitter of the end user (e.g., upstream transmitter **160**, FIG. **1**) as a seed signal for a modulator (e.g., modulator **164**, FIG. **1**) for upstream data transmission, as explained further below with respect to FIG. **9**.

(61) FIG. **9** is a flow chart diagram of an exemplary upstream optical network process **900** that can be optionally implemented with fiber communication system **100**, depicted in FIG. **1**. Process **900** begins at optional step **902**. In step **902**, where polarization division multiplexing is utilized in the upstream transmitter (e.g., upstream transmitter **160**, FIG. **1**), second unmodulated signal **170** (Ch1') (from step **826**, FIG. **8**) is split into its x-component and y-component parts (e.g., by PBS/PBC **604**, FIG. **6**) for separate direct or external modulation. Where polarization division multiplexing is not utilized, step **902** is skipped, and process **900** instead begins at step **904**.

(62) In step **904**, second unmodulated signal **170** (Ch1'), or its polarized components if optional step **902** is implemented, is injection locked to the master source laser (e.g., external laser **118**, FIG. **1**), as described above with respect to FIGS. **1** and **6**. Step **904** then proceeds to step **906**,

where injection locked signal is modulated by direct or external modulation. Process **900** then proceeds from step **906** to step **908**. Step **908** is an optional step, which is implemented if optional step **902** is also implemented for polarization division multiplexing. In step **908**, the x-component and y-component parts of the excited Ch1' signal are recombined (e.g., by PBS/PBC **604**, FIG. **6**) for output as upstream modulated data stream **176** (Ch1'). Where polarization division multiplexing was not utilized, process **900** skips step **908**, and instead proceeds directly from step **906** to step **910**.

(63) In step **910**, upstream modulated data stream **176** (Ch1') is optically multiplexed, i.e., by node optical multiplexer **178**, FIG. **1**, with other upstream data stream signals (not shown). Process **900** then proceeds from step **910** to step **912**, where upstream modulated data stream **176** (Ch1') is transmitted over an optical fiber, i.e., upstream fiber **110**, FIG. **1**. Process **900** then proceeds from step **912** to step **914**, where upstream modulated data stream **176** (Ch1') is optically demultiplexed, e.g., by second hub optical demultiplexer **134**, which separates the selected data stream from the other upstream data stream signals, for transmission to a particular upstream receiver tuned to receive the modulated data stream. Process **900** then proceeds from step **914** to step **916**, where both components (e.g., first unmodulated signal **168** (Ch1), FIG. **8**, and upstream modulated data stream **176** (Ch1')) of the upstream data stream pair, e.g., first data stream pair **700**, FIG. **7**, are received by an upstream receiver (e.g., upstream receiver and 32, FIG. **1**) for heterodyne coherent detection.

(64) As illustrated in the exemplary embodiment, a difference between upstream and downstream signal transmission is that an entire synchronized modulated/unmodulated channel pair (e.g., second data stream pair **702**, FIG. **7**) can be transmitted in the downstream direction, whereas, in the upstream direction, only a data modulated signal (e.g., upstream modulated data stream **176** (Ch1')) to be transmitted over the upstream fiber connection, i.e., upstream fiber **110**. An advantage of the present configuration is that the LO for upstream coherent detection (e.g., at upstream receiver **132**, FIG. **1**) comes directly from the split signal, i.e., first unmodulated signal **168** (Ch1) generated from optical frequency comb generator **114** within optical hub **102**, after separation by first hub optical demultiplexer **124**, as depicted in FIG. **1**. Conventional systems typically require LO generation at each stage of the respective system. According to the present disclosure, on the other hand, relatively inexpensive slave lasers can be implemented throughout the system architecture for modulation and polarization multiplexing in both optical hub **102** and end user **106** components, without requiring an additional LO source at the end user.

(65) According to the present disclosure, utilization of dual-polarization optical transmitters, and by direct modulation of semiconductor lasers with coherent detection, is particularly beneficial for not only longhaul applications, but also for shortreach applications to reduce the cost of electronic hardware, while also rendering the overall network system architecture more compact. The present systems and methods further solve the conventional problem of synchronizing two laser sources over a long period of time. Utilization of the phase synchronized data stream pairs and slave lasers herein allows continual synchronization of the various laser sources throughout the system during its entire operation. These solutions can be implemented within coherent DWDM-PON system architectures for access networks in a cost-efficient manner.

(66) Utilization of the high quality optical comb source at the front end of the system thus further allows a plurality of simultaneous narrow bandwidth wavelength channels to be generated with easily controlled spacing, and therefore also simplified tuning of the entire wavelength comb. This centralized comb light source in the optical hub provides master seeding sources and LO signals that can be reused throughout the system, and for both downstream and upstream transmission. The implementation of optical injection, as described herein, further improves the performance of low-cost multi-longitudinal slave laser sources in terms of spectral bandwidth and noise properties. Access networks according to the present systems and methods thus achieve more efficient transmission of wavelengths through optical fibers, thereby increasing the capacity of transmitted

data, but at lower power, increased sensitivity, lower hardware cost, and a reduction in dispersion, DSP compensation, and error correction.

(67) Exemplary embodiments of fiber communication systems and methods are described above in detail. The systems and methods of this disclosure though, are not limited to only the specific embodiments described herein, but rather, the components and/or steps of their implementation may be utilized independently and separately from other components and/or steps described herein. Additionally, the exemplary embodiments can be implemented and utilized in connection with other access networks utilizing fiber and coaxial transmission at the end user stage.

(68) This written description uses examples to disclose the embodiments, including the best mode, and also to enable any person skilled in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

(69) Although specific features of various embodiments of the disclosure may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the disclosure, a particular feature shown in a drawing may be referenced and/or claimed in combination with features of the other drawings. For example, the following list of example claims represents only some of the potential combinations of elements possible from the systems and methods described herein.

(70) a(i). An injection locked transmitter for an optical communication network, comprising: a master seed laser source input substantially confined to a single longitudinal mode; an input data stream; and a laser injected modulator including at least one slave laser having a resonator frequency being injection locked to a frequency of the single longitudinal mode of the master seed laser source, wherein the laser injected modulator is configured to receive the master seed laser source input and the input data stream, and output a laser modulated data stream.

(71) b(i). The transmitter of claim a(i), wherein the laser injected modulator is configured to implement direct modulation.

(72) c(i). The transmitter of claim a(i), wherein the laser injected modulator is configured to implement external modulation.

(73) d(i). The transmitter of claim a(i), wherein the at least one slave laser comprises at least one of an LED, a Fabry Perot laser diode, and a vertical-cavity surface-emitting laser.

(74) e(i). The transmitter of claim a(i), further comprising a first optical circulator in communication with the laser injected modulator and the master seed laser source input.

(75) f(i). The transmitter of claim e(i), wherein the laser injected modulator is configured to implement one of polarization division multiplexing, space division multiplexing, and mode division multiplexing.

(76) g(i). The transmitter of claim f(i), wherein the laser injected modulator is configured to multiplex the master seed laser source input at one of a 90 degree polarization, a 60 degree polarization, a 90 degree polarization, a spiral polarization, a circular polarization, a vortex polarization, or an orbital angular momentum.

(77) h(i). The transmitter of claim f(i), further comprising a polarization beam splitter and a polarization beam combiner disposed between the first optical circulator and the at least one slave laser.

(78) i(i). The transmitter of claim h(i), wherein the at least one slave laser comprises a first laser diode and a second laser diode, wherein the first laser diode is configured to receive an x-component of the master seed laser source input, and wherein the second laser diode is configured to receive a y-component of the master seed laser source input.

(79) j(i). The transmitter of claim i(i), further comprising: a first optical splitter and optical

combiner disposed between the polarization beam splitter and the first laser diode; and a second optical splitter and optical combiner disposed between the polarization beam splitter and the second laser diode, wherein the first laser diode comprises a first sub-laser and a second sub-laser, wherein the second laser diode comprises a third sub-laser and a fourth sub-laser, wherein the first sub-laser is configured to receive an I-signal of the x-component, wherein the second sub-laser is configured to receive a Q-signal of the x-component, wherein the third sub-laser is configured to receive an I-signal of the y-component, wherein the fourth sub-laser is configured to receive a Q-signal of the y-component.

(80) k(i). The transmitter of claim j(i), further comprising: a first phase shift element disposed between the first optical splitter and the second sub-laser; and a second phase shift element disposed between the second optical splitter and the fourth sub-laser.

(81) l(i). The transmitter of claim e(i), further comprising: a second optical circulator in one-way communication with the first optical circulator and in two-way communication with the at least one slave laser; and an external modulation element disposed between the first optical circulator and the second optical circulator, wherein the external modulation element is configured to receive the input data stream and an output of the second optical circulator, wherein the first optical circulator is in one-way communication with an output of the external modulation element.

(82) a(ii). An optical network communication system, comprising: an input signal source; an optical frequency comb generator configured to receive the input signal source and output a plurality of phase synchronized coherent tone pairs, each of the plurality of phase synchronized coherent tone pairs including a first unmodulated signal and a second unmodulated signal, a first transmitter configured to receive the first unmodulated signal of a selected one of the plurality of phase synchronized coherent tone pairs as a seed source and to output a first modulated data stream; and a first receiver configured to receive the first modulated data stream from the first transmitter and receive the second unmodulated signal of the selected one of the plurality of phase synchronized coherent tone pairs as a local oscillator source.

(83) b(ii). The system of claim a(ii), wherein the optical frequency comb generator comprises an amplifier and an optical demultiplexer.

(84) c(ii). The system of claim a(ii), wherein the optical frequency comb generator is configured to implement one of a mode-locked laser, a gain-switched laser, and electro-optical modulation.

(85) d(ii). The system of claim a(ii), wherein the selected one of the plurality of phase synchronized coherent tone pairs are controllable at a constant frequency spacing with respect to one another throughout the system.

(86) e(ii). The system of claim a(ii), wherein the system is configured to perform heterodyne detection.

(87) f(ii). The system of claim a(ii), wherein the first transmitter comprises a first laser injected modulator and a first optical circulator.

(88) g(ii). The system of claim a(ii), wherein the first laser injected modulator is configured to implement direct modulation.

(89) h(ii). The system of claim a(ii), wherein the first laser injected modulator is configured to implement external modulation.

(90) i(ii). The system of claim g(ii), wherein the input signal source includes an external master laser.

(91) j(ii). The system of claim i(ii), wherein the first modulator comprises a first laser diode configured to injection lock to the external master laser.

(92) k(ii). The system of claim j(ii), wherein the first laser diode is configured to receive first data from a first external data source to adhere into the output first modulated data stream.

(93) l(ii). The system of claim j(ii), wherein the first modulator further comprises a first polarization beam splitter and a first polarization beam combiner.

(94) m(ii). The system of claim l(ii), wherein the first laser diode comprises first and second slave

lasers, wherein the first and second slave lasers are configured to receive first and second polarization components, respectively, from the first polarization beam splitter.

(95) n(ii). The system of claim m(ii), wherein the first modulator is configured to implement quadrature amplitude modulation.

(96) o(ii). The system of claim n(ii), wherein the first slave laser comprises a first sub-laser and a second sub-laser, wherein the second slave laser comprises a third sub-laser and a fourth sub-laser, wherein the first and second sub-lasers are configured to receive an I-signal and a Q-signal of the first polarization component, respectively, and wherein the third and fourth sub-lasers are configured to receive an I-signal and a Q-signal of the second polarization component, respectively.

(97) p(ii). The system of claim a(ii), further comprising a second transmitter configured to receive the second unmodulated signal of the selected one of the plurality of phase synchronized coherent tone pairs as a seed source and to output a second modulated data stream.

(98) q(ii). The system of claim p(ii), wherein the second transmitter is configured to implement one of direct and external modulation.

(99) r(ii). The system of claim p(ii), wherein the second transmitter is configured to implement one or more of polarization division multiplexing and quadrature amplitude modulation.

(100) s(ii). The system of claim p(ii), further comprising a second receiver configured to receive the second modulated data stream from the second transmitter and receive the first unmodulated signal of the selected one of the plurality of phase synchronized coherent tone pairs as a local oscillator source.

(101) a(iii). An optical network communication system, comprising: an optical hub including an optical frequency comb generator configured to output at least one phase synchronized coherent tone pair having a first unmodulated signal and a second unmodulated signal, and a downstream transmitter configured to receive the first unmodulated signal as a seed source and to output a downstream modulated data stream; a fiber node; and an end user including a downstream receiver configured to receive the downstream modulated data stream from the downstream transmitter and receive the second unmodulated signal as a local oscillator source.

(102) b(iii). The system of claim a(iii), wherein the selected one of the plurality of phase synchronized coherent tone pairs are controllable at a constant frequency spacing with respect to one another throughout the system.

(103) c(iii). The system of claim a(iii), wherein the optical hub further comprises an amplifier, a first hub optical demultiplexer, and a hub optical multiplexer.

(104) d(iii). The system of claim c(iii), wherein the fiber node comprises a node optical demultiplexer configured to demultiplex an output from the hub optical multiplexer.

(105) e(iii). The system of claim d(iii), wherein the hub optical multiplexer is configured to communicate with the node optical demultiplexer by way of a downstream fiber.

(106) f(iii). The system of claim d(iii), wherein the node optical demultiplexer is configured to communicate with the downstream transmitter by way of first fiber optics.

(107) g(iii). The system of claim a(iii), wherein the downstream receiver comprises a downstream integrated coherent receiver, a downstream analog to digital converter, and a downstream digital signal processor.

(108) h(iii). The system of claim f(iii), wherein the end user further comprises an upstream transmitter, wherein the fiber node further comprises a node optical multiplexer, and wherein the optical hub further comprises a second hub optical demultiplexer and an upstream receiver.

(109) i(iii). The system of claim h(iii), wherein the upstream transmitter is configured to communicate with the node optical multiplexer by way of second fiber optics, and wherein the node optical multiplexer is configured to communicate with the second hub optical demultiplexer by way of an upstream fiber.

(110) j(iii). The system of claim i(iii), wherein the upstream transmitter is configured to receive the second unmodulated signal as a seed source and to output a upstream modulated data stream to the

node optical multiplexer.

(111) k(iii). The system of claim i(iii), wherein the upstream receiver comprises an upstream integrated coherent receiver, an upstream analog to digital converter, and an upstream digital signal processor.

(112) l(iii). The system of claim i(iii), wherein the upstream receiver is configured to receive as a data source the upstream modulated data stream from the node optical multiplexer, and to receive from the first hub optical demultiplexer the first unmodulated signal as a local oscillator source.

(113) m(iii). The system of claim a(iii), wherein the at least one phase synchronized coherent tone pair is controllable at a constant frequency spacing with respect to one another throughout the system.

(114) n(iii). The system of claim a(iii), wherein the end user comprises at least one of a customer device, customer premises, a business user, and an optical network unit.

(115) o(iii). The system of claim a(iii), further configured to implement coherent dense wavelength division multiplexing passive optical network architecture.

(116) p(iii). The system of claim i(iii), wherein the downstream digital signal processor is configured to hold constant the difference in frequency separation ω between the second unmodulated signal and the downstream modulated data stream when calculating a digital down conversion compensation factor $e^{j\omega t}$.

(117) q(iii). The system of claim k(iii), wherein the upstream digital signal processor is configured to hold constant the difference in frequency separation ω between the first unmodulated signal and the upstream modulated data stream when calculating the digital down conversion compensation factor $e^{j\omega t}$.

(118) a(iv). A method of optical network processing, comprising the steps of: generating at least one pair of first and second unmodulated phase synchronized coherent tones; transmitting the first unmodulated phase synchronized coherent tone to a first transmitter as a seed signal; adhering downstream data, in the first transmitter, to the first unmodulated phase synchronized coherent tone to generate a first modulated data stream signal; optically multiplexing the first modulated data stream signal and the second unmodulated phase synchronized coherent tone together within a hub optical multiplexer; and communicating the multiplexed first modulated data stream signal and the second unmodulated phase synchronized coherent tone to a first receiver, by way of fiber optics, for downstream heterodyne detection.

(119) b(iv). The method of claim a(iv), further comprising, prior to the step of adhering downstream data, the step of polarization beam splitting the first unmodulated phase synchronized coherent tone.

(120) c(iv). The method of claim b(iv), further comprising, after the step of adhering downstream data, and prior to the step of optically multiplexing, the step of polarization beam combining split components from the step of polarization beam splitting of the first unmodulated phase synchronized coherent tone.

(121) d(iv). The method of claim a(iv), wherein the step of adhering downstream data implements injection locking.

(122) e(iv). The method of claim a(iv), further comprising the steps of: optically splitting, prior to the step of communicating, the second unmodulated phase synchronized coherent tone; and receiving, by a second transmitter, a portion of the optically split second unmodulated phase synchronized coherent tone as local oscillator for upstream detection.

(123) f(iv). The method of claim e(iv), further comprising a step of adhering upstream data, in the second transmitter, to the second unmodulated phase synchronized coherent tone to generate a second modulated data stream signal.

(124) g(iv). The method of claim f(iv), wherein the step of adhering upstream data comprises a step of injection locking a slave laser to an external master laser.

(125) h(iv). The method of claim f(iv), further comprising, prior to the step of adhering upstream

data, the step of polarization beam splitting the second unmodulated phase synchronized coherent tone.

(126) i(iv). The method of claim h(iv), further comprising, after the step of adhering upstream data, the step of polarization beam combining split components from the step of polarization beam splitting of the second unmodulated phase synchronized coherent tone.

(127) j(iv). The method of claim f(iv), further comprising a step of transmitting the second modulated data stream signal to a second receiver, by way of fiber optics, for upstream heterodyne detection.

(128) Some embodiments involve the use of one or more electronic or computing devices. Such devices typically include a processor or controller, such as a general purpose central processing unit (CPU), a graphics processing unit (GPU), a microcontroller, a reduced instruction set computer (RISC) processor, an application specific integrated circuit (ASIC), a programmable logic circuit (PLC), a field programmable gate array (FPGA), a DSP device, and/or any other circuit or processor capable of executing the functions described herein. The processes described herein may be encoded as executable instructions embodied in a computer readable medium, including, without limitation, a storage device and/or a memory device. Such instructions, when executed by a processor, cause the processor to perform at least a portion of the methods described herein. The above examples are exemplary only, and thus are not intended to limit in any way the definition and/or meaning of the term “processor.”

(129) This written description uses examples to disclose the embodiments, including the best mode, and also to enable any person skilled in the art to practice the embodiments, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. A transceiver for an optical communication network, comprising: a transceiver input configured to receive (i) a first coherent tone that is modulated with first data as a data-modulated first coherent tone, and (ii) a second coherent tone as an unmodulated second coherent tone, wherein the second coherent tone is phase synchronized with the first coherent tone; a downstream receiver in communication with the transceiver input, and configured to (i) coherently detect the data-modulated first coherent tone, and (ii) utilize the unmodulated second coherent tone as a local oscillator source; and an upstream transmitter configured to (i) modulate second data onto the second coherent tone as a data-modulated second coherent tone, and (ii) transmit upstream the data-modulated second coherent tone along with an unmodulated first coherent tone, wherein a frequency difference between the first and second coherent tones is held constant throughout the optical communication network.
2. The transceiver of claim 1, further comprising an optical splitter disposed between the transceiver input and the downstream receiver.
3. The transceiver of claim 2, wherein the optical splitter is configured to split the unmodulated second coherent tone into at least first and second portions of the unmodulated second coherent tone, and wherein the downstream receiver is further configured to receive the first portion of the second coherent tone as the local oscillator source.
4. The transceiver of claim 3, wherein the upstream transmitter is in communication with the optical splitter and configured to receive the second portion of the unmodulated second coherent tone prior to adhering the second data.
5. The transceiver of claim 4, wherein the upstream transmitter further comprises a modulator (i) in

- communication with the optical splitter, and (ii) configured to adhere the second data as upstream data onto the second portion of the unmodulated second coherent tone to generate the data-modulated second coherent tone as a modulated upstream data stream signal.
6. The transceiver of claim 5, further comprising a transmitter output in operable communication with the modulator, and configured to communicate the modulated upstream data stream signal to an upstream receiver, over an optical communication medium in operable communication with the transmitter output, for upstream coherent detection by the upstream receiver.
7. The transceiver of claim 6, wherein the transmitter output comprises an optical circulator disposed between the modulator and the transceiver input.
8. The transceiver of claim 6, wherein the upstream receiver is configured for upstream heterodyne detection using the unmodulated first coherent tone as a local oscillator source.
9. The transceiver of claim 1, wherein the downstream receiver is further configured for downstream heterodyne detection.
10. The transceiver of claim 1, wherein the frequency difference between the first and second coherent tones is held constant according to a compensation factor $e^{j\omega t}$, where t represents a time parameter, and where ω represents the frequency difference that is held constant between the first and second coherent tones.
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